Link for the GitHub video so you can watch it on your own:
https://youtube.com/watch?v=w3jLJU7DT5E

Reminders

Upcoming due dates
Nothing due this week

Discussion section this week is a tour of Datahub and an ungraded Python review notebook

Version Control 2.0

Data Science in Practice

Jason G. Fleischer, PhD Dept. of Cognitive Science UC San Diego https://jgfleischer.com

Each time you create a commit, git tracks the changes made automatically.

By committing each time you make changes, git allows you to time travel!

By committing each time you make changes, git allows you to time travel!

377dfcd00dd057542b112cf13be6cf1380b292ad

439301fe69e8f875c049ad0718386516b4878e22

456722223e9f9e0ee0a92917ba80163028d89251

There's a unique id, known as a **hash**, associated with each commit.

You can return to the state of the repository at any commit. Future commits don't disappear. They just aren't visible when you **check out** an older commit.

377dfcd00dd057542b112cf13be6cf1380b292ad

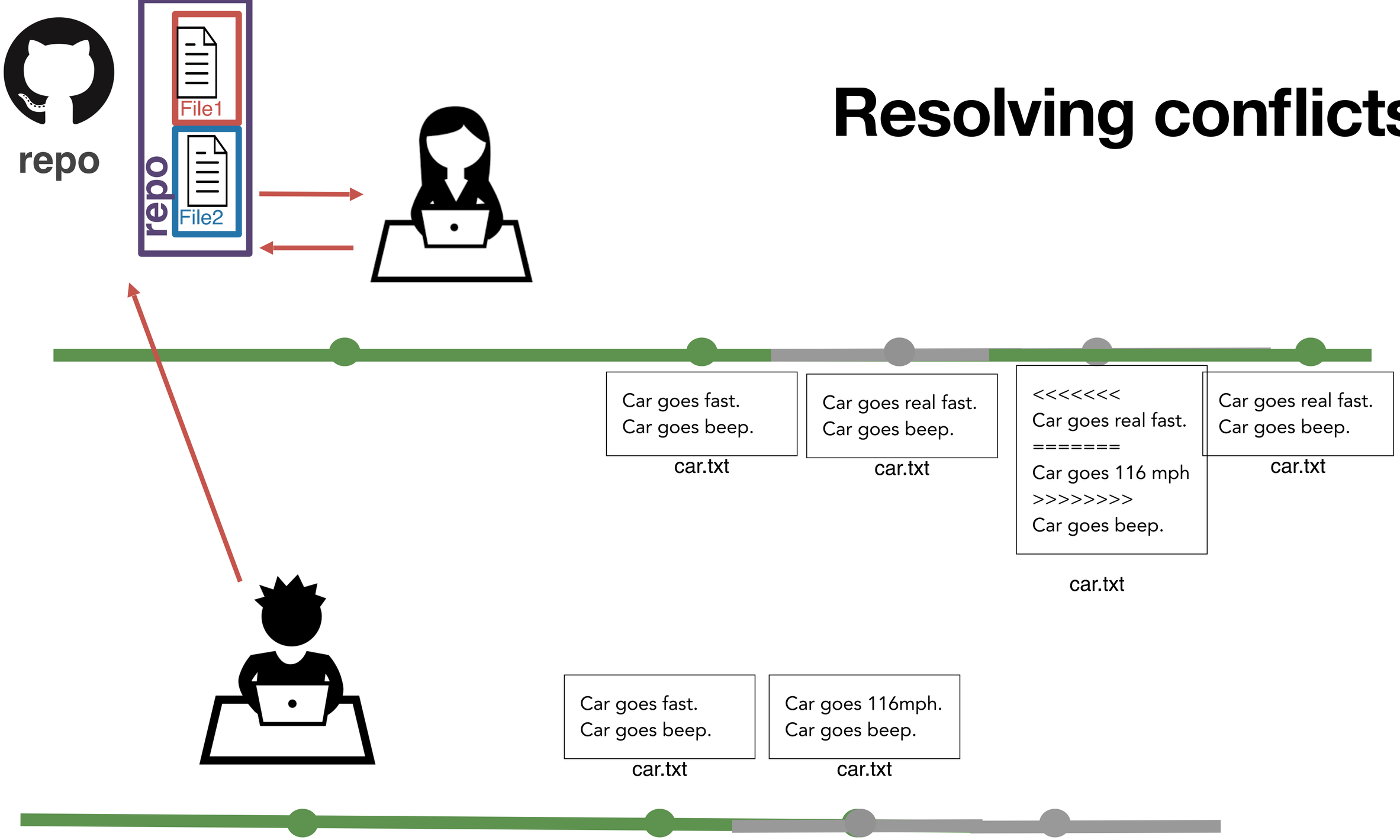
```
(base) jasonfleischer@rabona project_managment % git log
commit 08a2877e4ccd1f5440b7364feb7add64e4c926f9 (HEAD -> main, origin/main, origin/HEAD)
Author: Jason Fleischer <jason.g.fleischer@gmail.com>
Date:   Fri Oct 17 22:02:17 2025 -0700

    new style final project rubric

commit e8e9e5cd34ac494d06e97a7e494fd6c929bf3ff0
Author: Jason Fleischer <jason.g.fleischer@gmail.com>
Date:   Mon Apr 21 21:34:54 2025 -0700

    Update README.md

commit 4a7e23dab836f9b743af3ffc3ed0892561be216d
Author: Shannon Ellis <shannon0ellis@gmail.com>
Date:   Wed Jan 1 14:23:03 2025 -0800

    WI25; added note about proposal to data checkpoint points returned
```

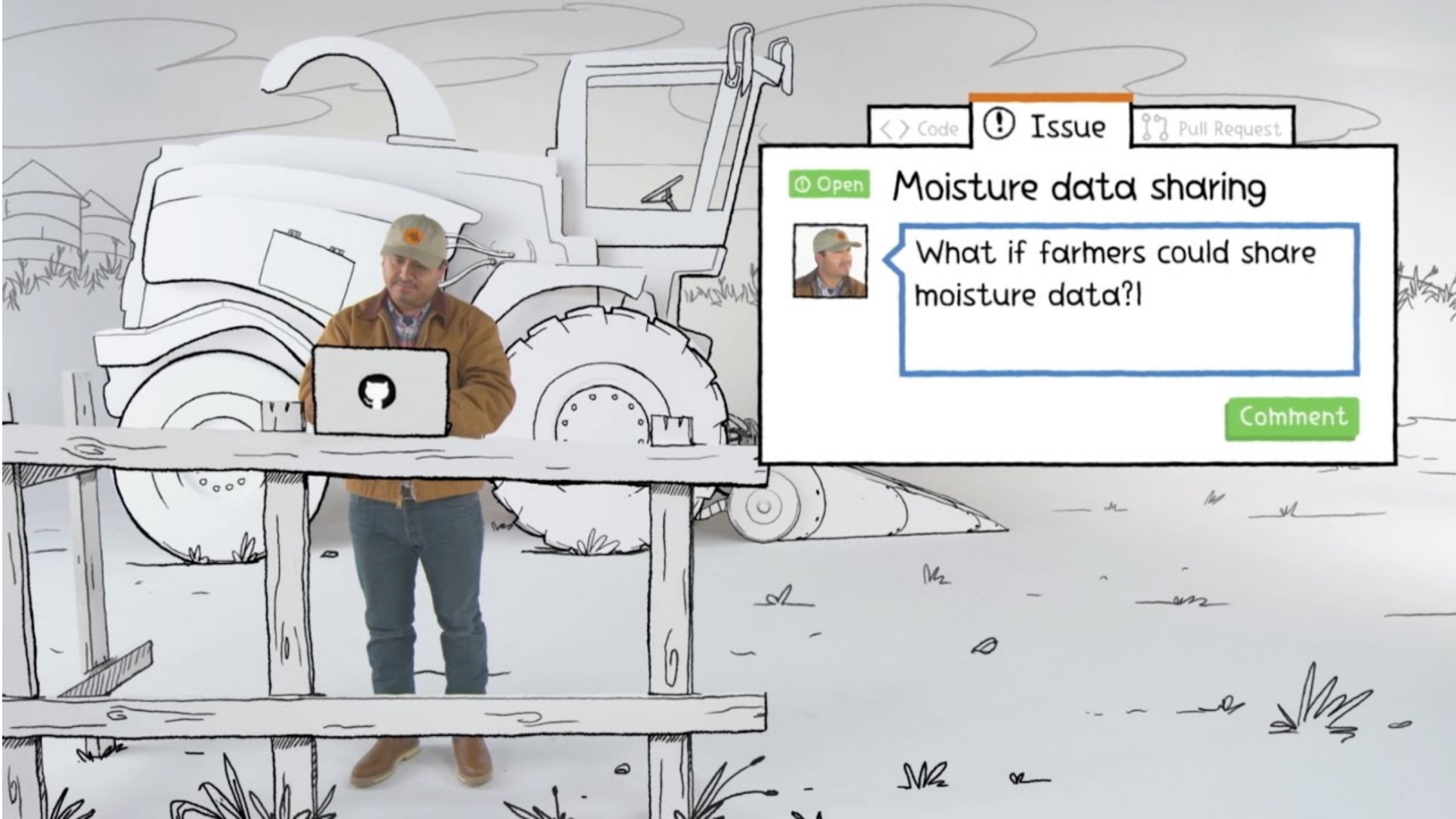
Use git log
to find the hash…

```
(base) jasonfleischer@rabona project_managment % git log --oneline
08a2877 (HEAD -> main, origin/main, origin/HEAD) new style final project rubric
e8e9e5c Update README.md
4a7e23d WI25; added note about proposal to data checkpoint points returned
587b422 modified feedback
f343df9 Tried to make rubric more distinguishing of proficient work
f4d6771 removed abstract which wasn't in the template
1ebf4a7 Update sync_project_grading_from_GitHub_to_Canvas.ipynb
3677607 Create Final Project Grading Rubric
088eb11 added EDA
66e5aa6 re-arrange Grading file
d022916 update data checkpoint grading rubrics
9e8c637 creates the data checkpoint feedback template
cf8d4ae Update README.md
f7f3a03 Sync Score Notebook
818a832 Update README.md
5bacac0 FA23 Project Creation and Grading
6524577 Initial commit
```

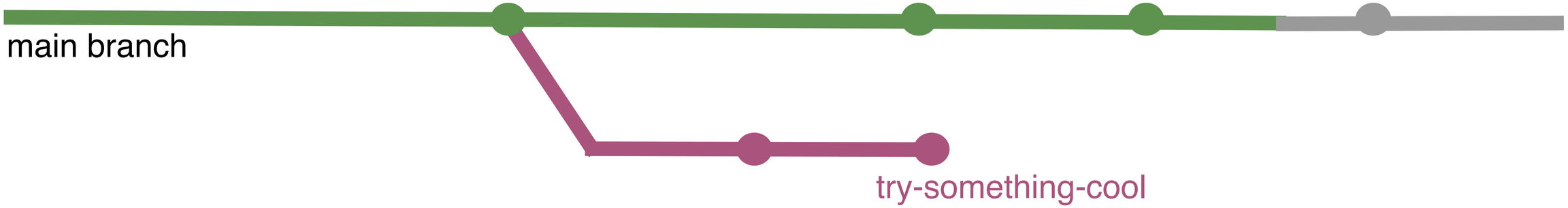
And I'll do a little
live demo of checkout
in our current lecture repo

Resolving conflicts

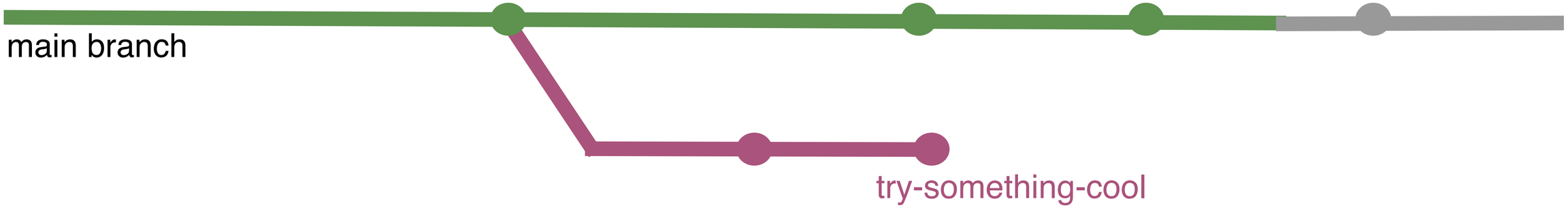



Code
Issue
Pull Request
Open
Moisture data sharing
What if farmers could share moisture data?!
Comment

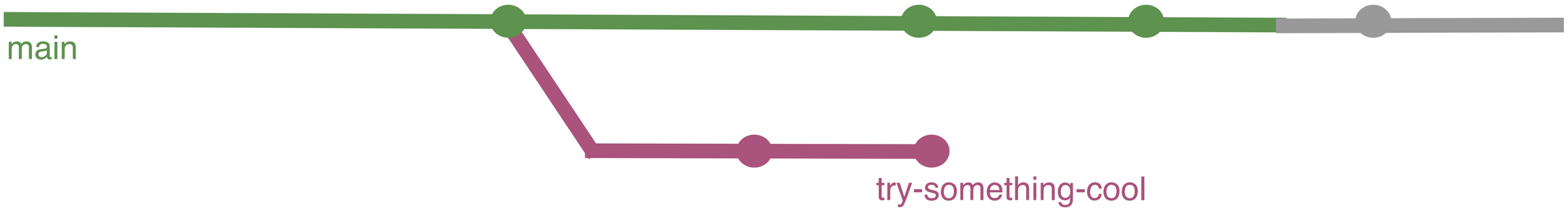
But...not everything is always linear. Sometimes you want to try something out and you're not sure it's going to work. This is where you'll want to use a **branch**.

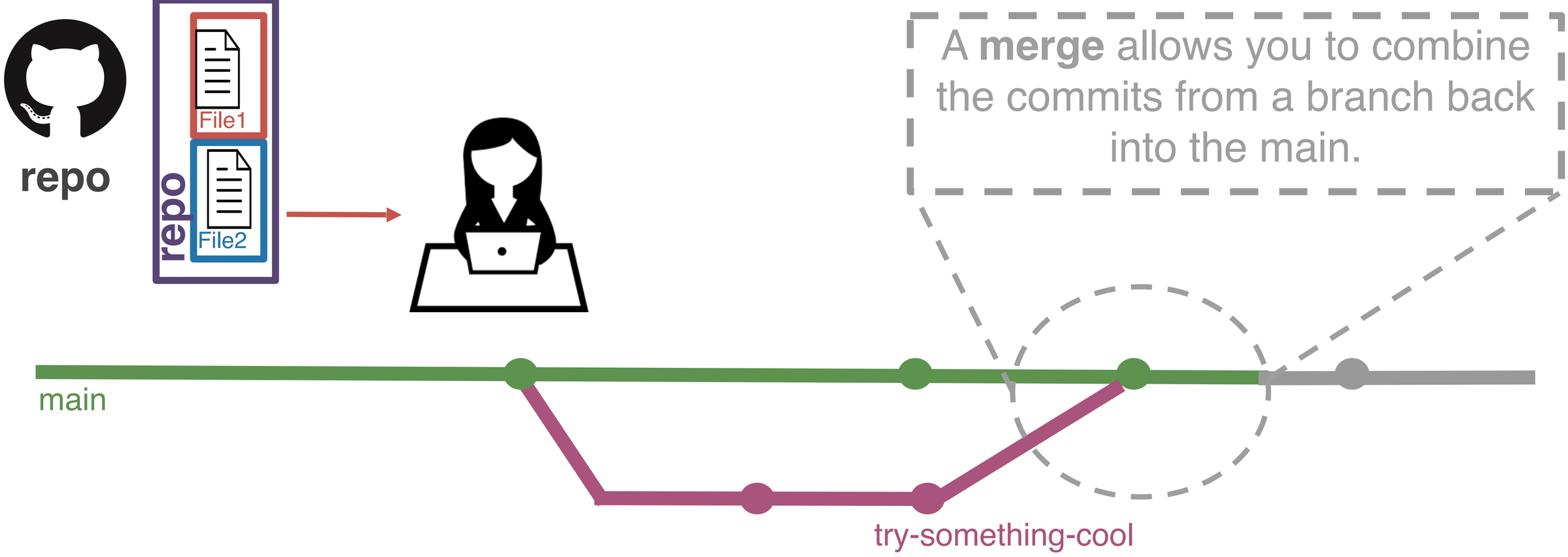


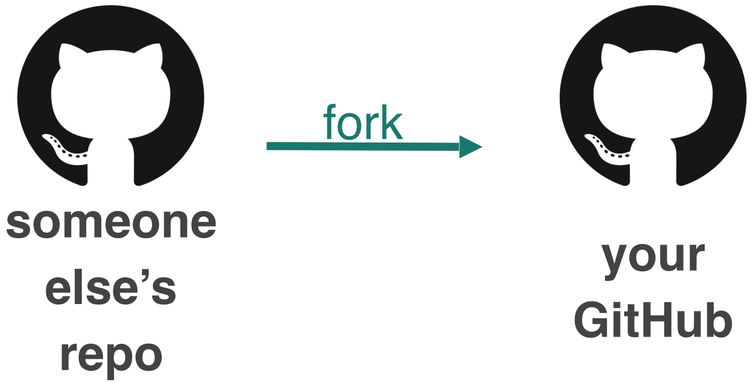
It's a good way to experiment. It's pretty easy to get rid of a branch later on should you not want to include the commits on that branch.



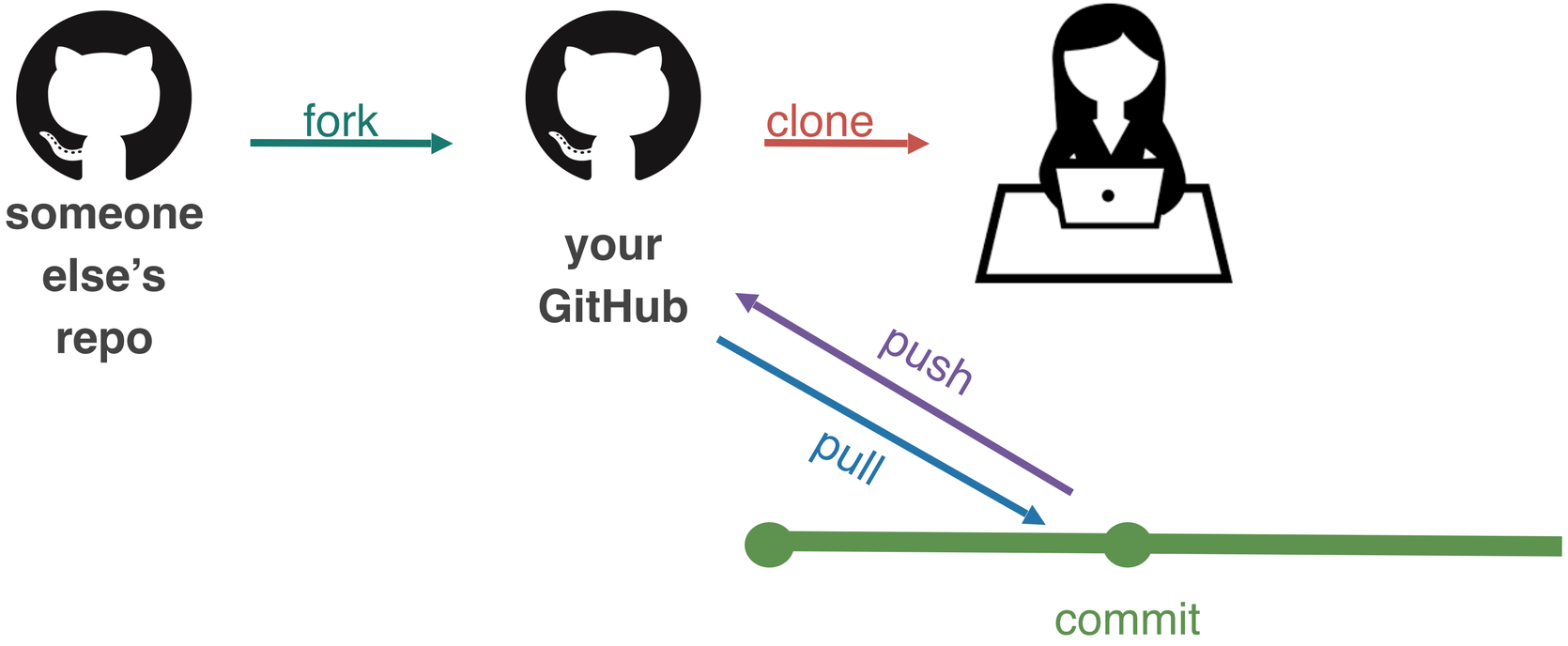
But...what if you DO want to include the changes you've made on your try-something-cool branch into the main branch?



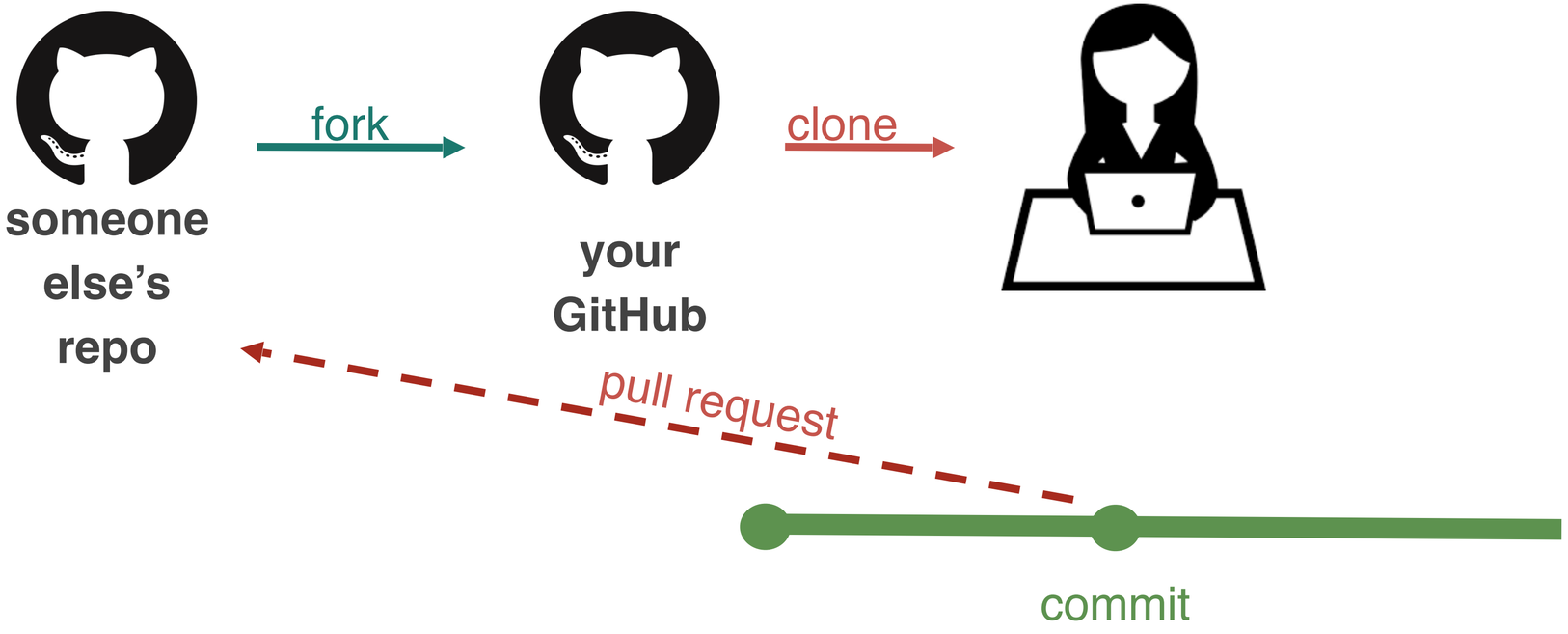

repo
repo
File1
File2
A **merge** allows you to combine the commits from a branch back into the main.
main
try-something-cool



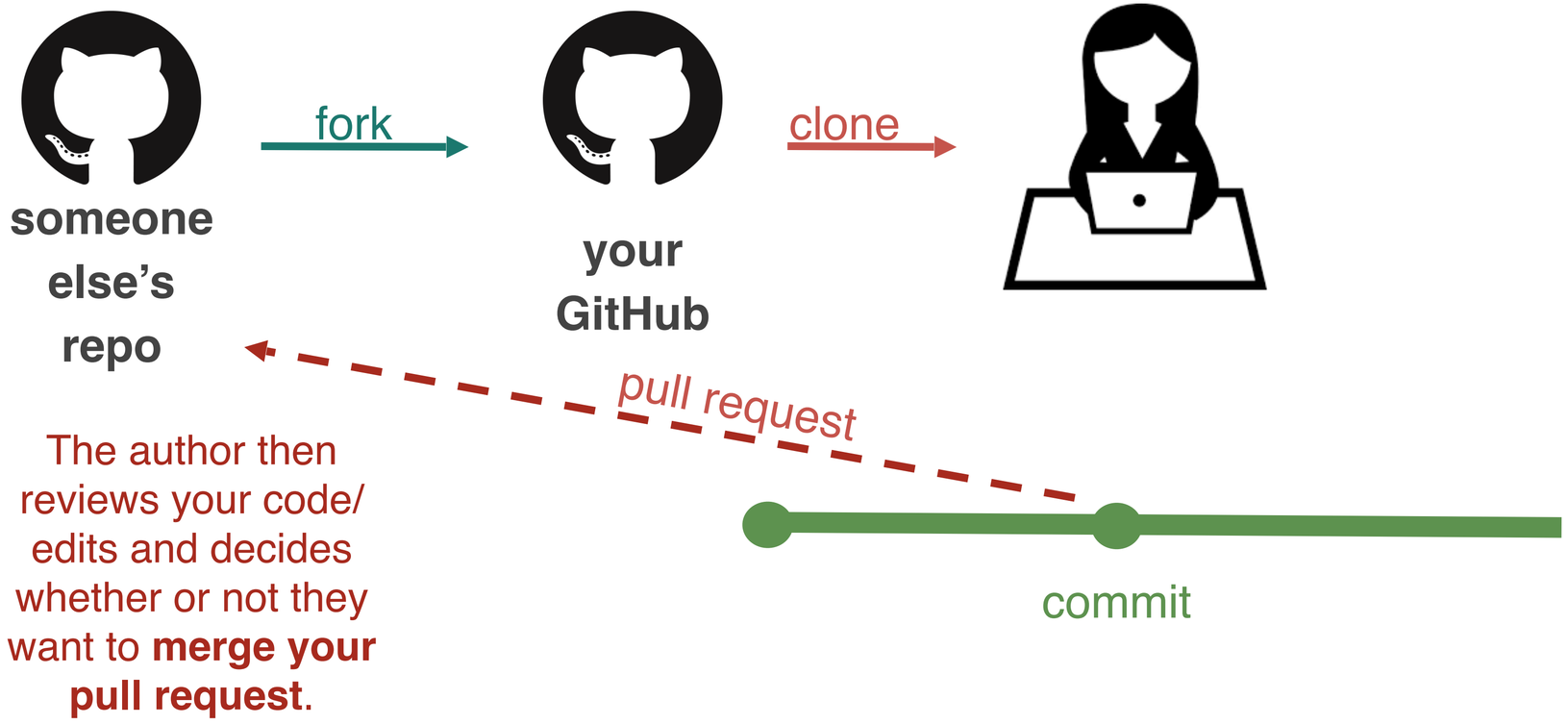
What if someone else is working on something cool and you want to play around with it? You'll have to **fork** their repo.



After you fork their repo, you can play around with it however you want, using the workflow we've already discussed.



But what if you think you've found a bug in their code, a typo, or want to add a new feature to their software? For this, you'll submit a **pull request** (aka **PR**).



The author then reviews your code/ edits and decides whether or not they want to **merge your pull request**.

But what if you think you've found a bug in their code, a typo, or want to add a new feature to their software? For this, you'll submit a **pull request** (aka **PR**).

someone else's repo

Last but not least...what if you find a bug in someone else's code OR you want to make a suggestion but aren't going to submit a suggestion with a PR. For this, you can file an **issue** on GitHub.

someone else's repo

Last but not least...what if you find a bug in someone else's code OR you want to make a suggestion but aren't going to submit a suggestion with a PR. For this, you can file an **issue** on GitHub.

Issues are *bug trackers*. While, they can include bugs, they can also include feature requests, to-dos, whatever you want, really!

They can be assigned to people.

They can be closed once addressed ….or if the software maintainer doesn't like the suggestion

GitHub Issues

- Can also be used for team coordination like you saw in the video
- Will be used for your grade feedback!!

Research Question

How does UCSD's student engagement and social life compare to that of other UC campuses and do these findings support the online narratives of UCSD being "socially dead" during 2020 to 2024?

To answer this question we are planning on measuring student engagement trends in each respective campus through data on student organizations and student events hosted. This data will be collected through reddit, social calendars on the campus website, campus websites on school organizations, and social media engagement from campuses or student organizations. Through correlation methods such as regression lines and r correlation, ANOVA tests, geospatial analysis, and statistical correlation analysis the data will be visualized to approve or disapprove the hypothesis.

Background and Prior Work

For many years, the University of California, San Diego (UCSD) has been referred to as "UC Socially Dead" in popular media, online discussions, and social commentary, developing a reputation for having a less vibrant social scene compared to other universities. The "Socially Dead" narrative has been both defended and challenged by students and alumni, shaping the expectations of prospective students and influencing yearly college commitments. While some argue that UCSD offers a thriving social environment through clubs, organizations, and campus events, others claim that the school falls short in these areas, resulting in a generally antisocial student body and a striking lack of school spirit. This study aims to explore and understand the quantifiable data that may underlie these stereotypes, using other UC campuses as semi-controlled points of comparison.

COGS108 / Group042_FA25

Code | Issues 5 | Pull requests | Actions | Projects | Wiki | Security | Insights | Settings

Project Proposal Feedback #1

Open

grader-cogs108 opened on Oct 27, 2025 · edited by JasonC1217 Edits

Project Proposal Feedback

Score (out of 10 pts)

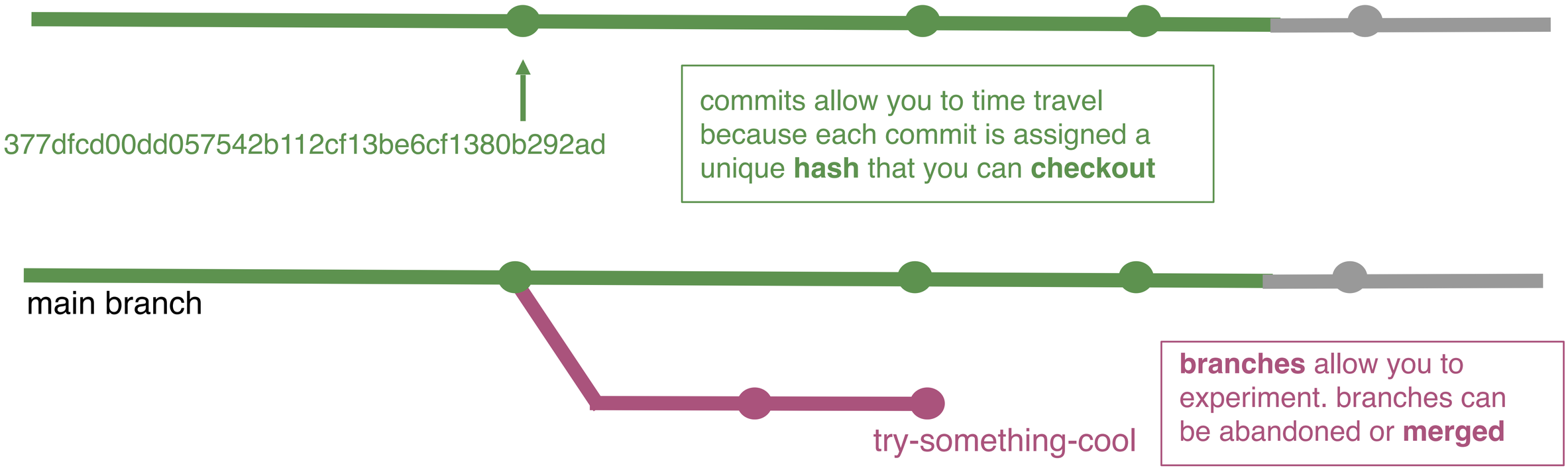
Score = 9

Feedback:

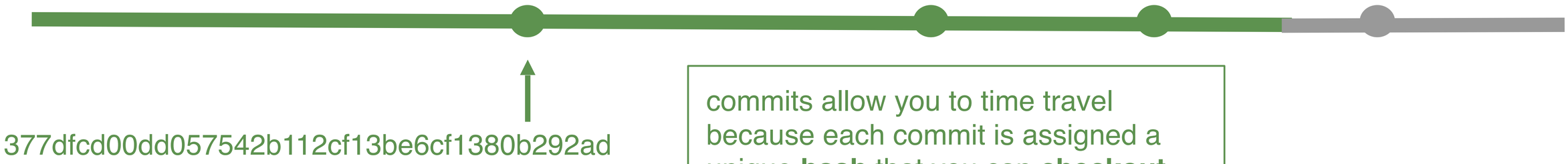
	Quality	Reasons
Research question	Proficient	Great research question, but key elements are under-specified: what exactly is "student engagement," which primary outcome will decide "socially dead," what unit of analysis (campus-year), and what controls (enrollment size, % commuters, housing capacity, COVID policy intensity) will be included. **Update**: Addressed feedback on research question.
Background	Proficient	You cite a NORC methodology review and journalistic/Medium pieces; useful context, but limited peer-reviewed, UC-specific empirical work on engagement metrics and their validity. **Update**: Addressed feedback on the background.

commits allow you to time travel because each commit is assigned a unique **hash** that you can **checkout**

One more git recap...



One more git recap...



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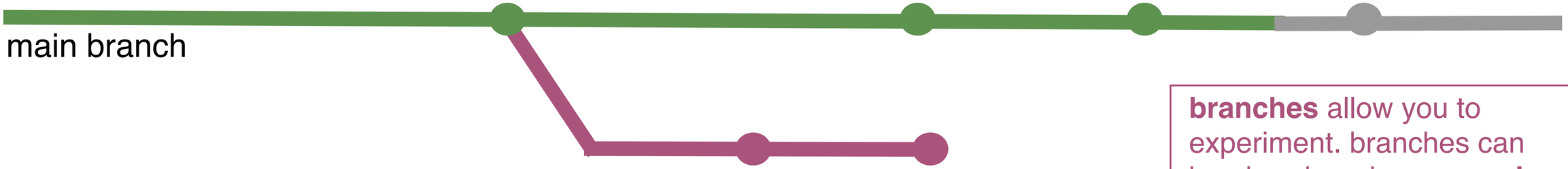
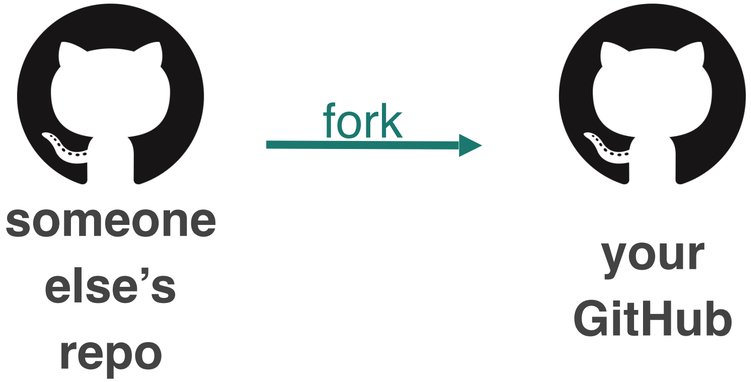
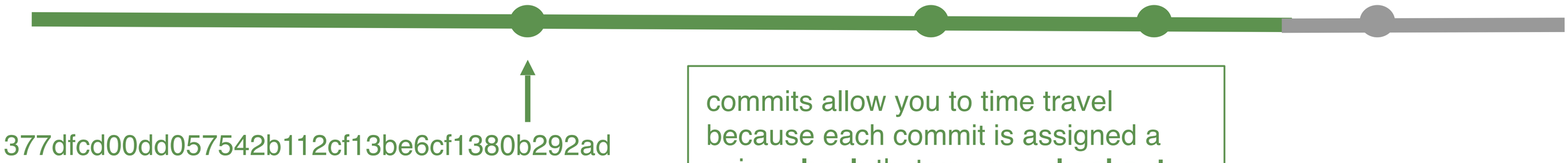


branches allow you to experiment. branches can be abandoned or **merged**



You can work on others' repos by first **forking** their repository onto your GitHub

One more git recap...



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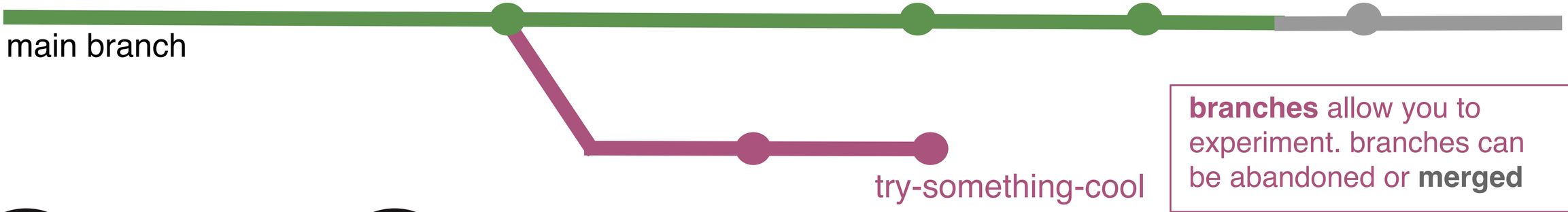
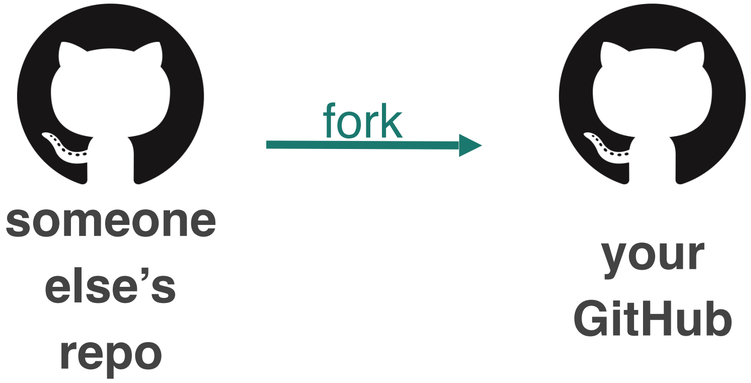


branches allow you to experiment. branches can be abandoned or **merged**



You can work on others' repos by first **forking** their repository onto your GitHub

Pull requests allow you to make specific edits to others' repos

Issues allow you to make general suggestions to your/others' repos

One more git recap...

Working with others in git

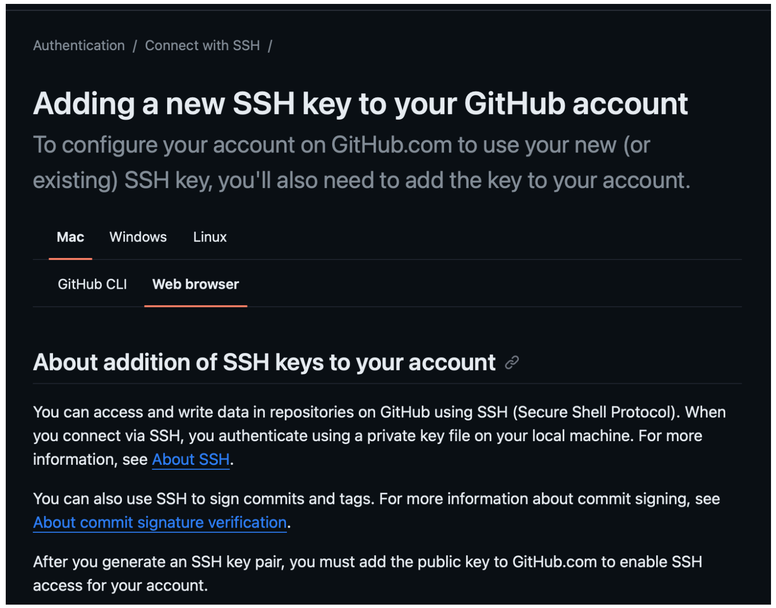
https://forms.gle/eyxgHB3wvqmy17uR9

Using SSH to contribute work to your project repo

You can’t write or access private repos without authentication!

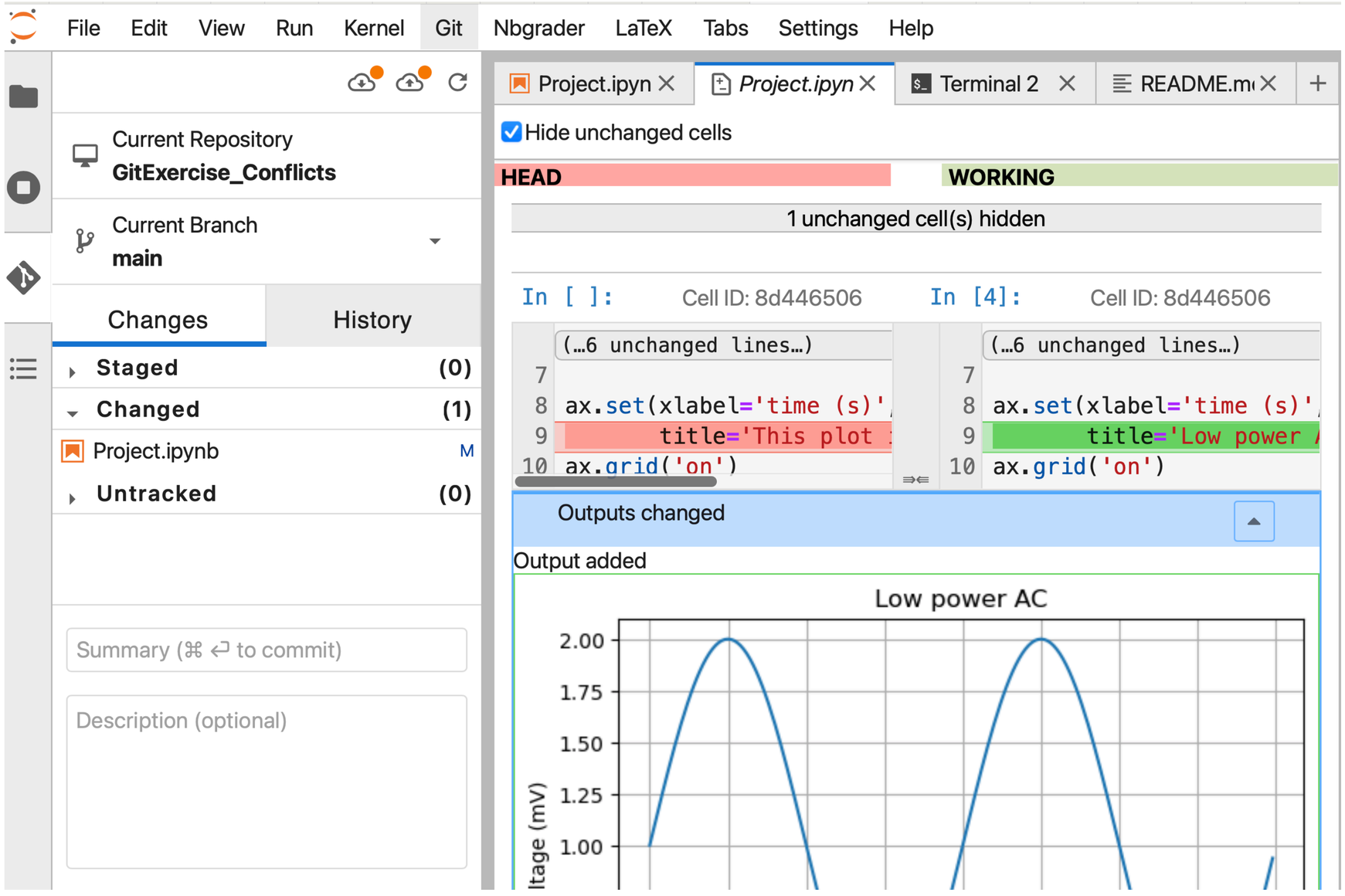
Click here, follow instructions:

https://docs.github.com/en/authentication/connecting-to-github-with-ssh/adding-a-new-ssh-key-to-your-github-account



You will need to do this for each local computer you will use
(e.g., on Datahub and again on your laptop!)

What version control looks like on notebooks



Datahub
@ UCSD

Jupyter notebooks suck to version control

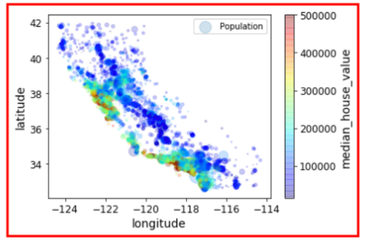
https://nextjournal.com/schmudde/how-to-version-control-jupyter

ReviewNB

ReviewNB is a GitHub app that also offers visual diffing with an interface that looks similar to the traditional Jupyter IDE. Because the outputs are visualized, problems associated with committing binary blobs disappear.

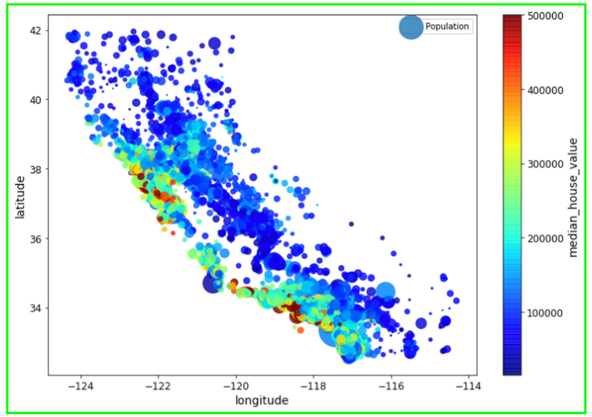
```
housing.plot(kind="scatter", x="longitude", y="latitude", alpha=0.2,
    s=housing["population"]/88, label="Population", figsize=(6,4),
    c="median_house_value", cmap=plt.get_cmap("jet"), colorbar=True,
    sharex=False)
plt.legend()
save_fig("housing_prices_scatterplot")
```

```
Saving figure housing_prices_scatterplot
```




```
housing.plot(kind="scatter", x="longitude", y="latitude", alpha=0.8,
    s=housing["population"]/21, label="Population", figsize=(10,7),
    c="median_house_value", cmap=plt.get_cmap("jet"), colorbar=True,
    sharex=False)
plt.legend()
save_fig("housing_prices_scatterplot")
```

```
Saving figure housing_prices_scatterplot
```



ReviewNB example courtesy of the ReviewNB website

😎 Good to Git

Use it for…

- Plain text files
- < 100MB per file limit
- Total repository size < 5 GB

⚠️ Careful

Easy to mess up

- Metadata rich text files that change over time, e.g.
  - Jupyter Notebook .ipynb
  - Word processor formats like .docx
  - JSON data

🙅‍♀️ Don't Git

Avoid using for

- Large text files
  ❌ 648MB .

✅ **Git LFS can help!**

- Vast directory structures
  ❌ 36,142,087 small text files in a nested subdirectory tree up to 14 levels deep
- Binary files
  ❌ .jpg, .mp4, .sql, etc

Lets practice!

At least if we have enough time right now :)

- Create repo : “mkdir some_new_repo; cd some_new_repo; git init”
- Create file.md: use an editor, e.g. a CL editor like “nano file.md”… ctrl+x to exit nano editor!
- Create a branch “my_work”: “git branch my_work; git checkout my_work”
- On “my_work”: Add a new file, prog.py using an editor
- On “my_work”: Change contents of file.md by adding newlines, save the file!
- Switch to “master” and merge “my_work” into “master” (problem free)
- On “master”: Change contents of file.md header
- On “my_work”: Change contents of file.md header in a way that conflicts with master
- Switch to “master” and merge “my_work”
- Resolve conflicts, finish merge

More practice

For outside of lecture

You need to learn by doing. Listening to me blah blah is only going to go so far. Here's some choices that seem to me like they would be good for beginner to intermediate levels, but note I have not actually used them.

- Free 16 hour Coursera course by Google covering common Git usage patterns https://www.coursera.org/learn/introduction-git-github
- Katas for Git https://github.com/eficode-academy/git-katas

- If you have other suggestions please let me know!
- If you use these and like or hate them please give me feedback!

Things you may wish to install on your laptop

Git (if not already installed!)
https://git-scm.com/downloads

Python and Jupyter via Anaconda
https://www.anaconda.com/download

NBDime is how to get ReviewNB-style diffs
in your laptop's Jupyter notebook
https://nbdime.readthedocs.io/en/latest/installing.html

Other things to consider according to your preferences:

- GitHub Desktop, Sourcetree, or some other GUI?
- VSCode and GitHub extensions?